CEUE203
25IT037

PRACTICAL – 12

Program 1: Word Frequency Using HashMap

Aim:

To implement word frequency counting using HashMap in Java and sort words by their frequency in descending order using Comparator.

Code:

```
import java.util.*;


public class WordFrequency {
    public static void main(String[] args) {


        String paragraph = "java is easy java is powerful java";


        String[] words = paragraph.toLowerCase().split("\\s+");


        HashMap<String, Integer> map = new HashMap<>();


        for (String word : words) {
            map.put(word, map.getOrDefault(word, 0) + 1);
        }


        List<Map.Entry<String, Integer>> list =
                new ArrayList<>(map.entrySet());
```

```
        list.sort((a, b) ->
                b.getValue().compareTo(a.getValue()));

        System.out.println("Word Frequency:");

        for (Map.Entry<String, Integer> entry : list) {
            System.out.println(entry.getKey() + " : "
                    + entry.getValue());
        }
    }
}
```

Output:

Word Frequency:

java : 3

is : 2

easy : 1

powerful : 1

Program 2: Unique Visitors Using HashSet and LinkedHashSet

Aim:

To implement unique visitor counting using HashSet and LinkedHashSet in Java and display the unique count and first three distinct visitors in insertion order.

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Code:

```
import java.util.*;


public class UniqueVisitors {
    public static void main(String[] args) {

        List<String> visitors = Arrays.asList(
                "Tirth", "Meet", "Priya",
                "Tirth", "Neha", "Meet",
                "Karan", "Priya"
        );

        HashSet<String> uniqueUsers =
                new HashSet<>(visitors);

        LinkedHashSet<String> orderedUsers =
                new LinkedHashSet<>(visitors);

        System.out.println("Unique Visitors Count: "
                + uniqueUsers.size());

        System.out.println("First Three Unique Visitors:");

        int count = 0;

        for (String name : orderedUsers) {
            if (count == 3) {
                break;
```

```
        }


        System.out.println(name);
        count++;
      }
   }
}
```

Output:

Unique Visitors Count: 5

First Three Unique Visitors:

Tirth

Meet

Priya

Program 3: Leaderboard Using Map and Comparator

Aim:

To implement a leaderboard using Map in Java and display the top N players sorted by score in descending order, with player names in alphabetical order when scores are equal.

Code:

```
import java.util.*;


public class Leaderboard {
   public static void main(String[] args) {


      Map<String, Integer> scores = new HashMap<>();
```

```
        scores.put("Tirth", 90);
        scores.put("Meet", 85);
        scores.put("Priya", 95);
        scores.put("Neha", 90);
        scores.put("Karan", 80);

        int topN = 3;

        List<Map.Entry<String, Integer>> players =
                new ArrayList<>(scores.entrySet());

        players.sort((a, b) -> {

            int result =
                    b.getValue().compareTo(a.getValue());

            if (result == 0) {
                return a.getKey().compareTo(b.getKey());
            }

            return result;
        });

        System.out.println("Top " + topN + " Players:");

        for (int i = 0; i < Math.min(topN, players.size()); i++) {
```

```
            Map.Entry<String, Integer> player =
                players.get(i);


            System.out.println(player.getKey()
                + " : " + player.getValue());
        }
    }
}
```

Output:

Top 3 Players:

Priya : 95

Neha : 90

Tirth : 90